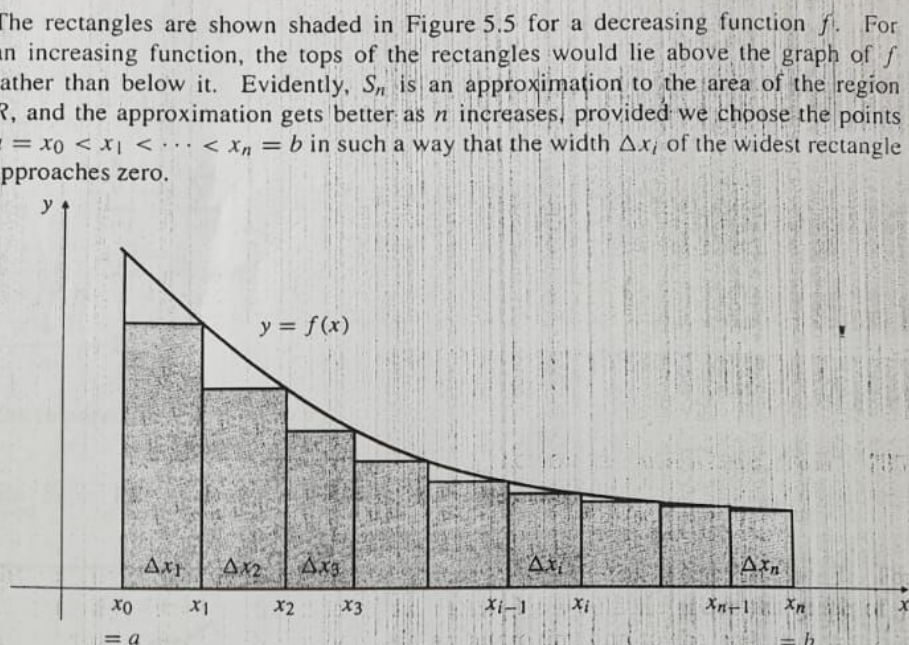


Figure 5.5 Approximating the area under the graph of a decreasing function using rectangles



Observe in Figure 5.6, for example, that subdividing a subinterval into two smaller subintervals reduces the error in the approximation by reducing that part of the area under the curve that is not contained in the rectangles. It is reasonable, therefore, to calculate the area of R by finding the limit of S_n as $n \rightarrow \infty$ with the restriction that the largest of the subinterval widths Δx_i must approach zero:

$$\text{Area of } R = \lim_{\substack{n \rightarrow \infty \\ \max \Delta x_i \rightarrow 0}} S_n.$$

Sometimes, but not always, it is useful to choose the points x_i ($0 \leq i \leq n$) in $[a, b]$ in such a way that the subinterval lengths Δx_i are all equal. In this case we have

$$\Delta x_i = \Delta x = \frac{b-a}{n}, \quad x_i = a + i \Delta x = a + \frac{i}{n}(b-a).$$

Some Area Calculations

We devote the rest of this section to some examples in which we apply the technique described above for finding areas under graphs of functions by approximating with rectangles. Let us begin with a region for which we already know the area so we can satisfy ourselves that the method does give the correct value.

EXAMPLE 1

Find the area A of the region lying under the straight line $y = x + 1$, above the x -axis and between the lines $x = 0$ and $x = 2$.

Solution The region is shaded in Figure 5.7(a). It is a *trapezoid* (a four-sided polygon with one pair of parallel sides) and has area 4 square units. (It can be divided into a rectangle and a triangle, each of area 2 square units.) We will calculate the area as a limit of sums of areas of rectangles constructed as described above. Divide the interval $[0, 2]$ into n subintervals of equal length by points

$$x_0 = 0, \quad x_1 = \frac{2}{n}, \quad x_2 = \frac{4}{n}, \quad x_3 = \frac{6}{n}, \quad \dots, \quad x_n = \frac{2n}{n} = 2.$$

The value of $y = x + 1$ at $x = x_i$ is $x_i + 1 = \frac{2i}{n} + 1$ and the i th subinterval, $\left[\frac{2(i-1)}{n}, \frac{2i}{n}\right]$, has length $\Delta x_i = \frac{2}{n}$. Observe that $\Delta x_i \rightarrow 0$ as $n \rightarrow \infty$. The sum

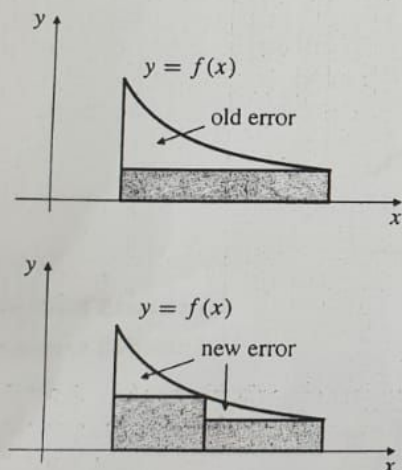


Figure 5.6 Using more rectangles makes the error smaller

of the areas of the approximating rectangles shown in Figure 5.7(a) is

$$\begin{aligned}
 S_n &= \sum_{i=1}^n \left(\frac{2i}{n} + 1 \right) \frac{2}{n} \\
 &= \left(\frac{2}{n} \right) \left[\frac{2}{n} \sum_{i=1}^n i + \sum_{i=1}^n 1 \right] \quad (\text{Use parts (b) and (a) of Theorem 1.}) \\
 &= \left(\frac{2}{n} \right) \left[\frac{2}{n} \frac{n(n+1)}{2} + n \right] \\
 &= 2 \frac{n+1}{n} + 2.
 \end{aligned}$$

Therefore, the required area A is given by

$$A = \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \left(2 \frac{n+1}{n} + 2 \right) = 2 + 2 = 4 \text{ square units.}$$

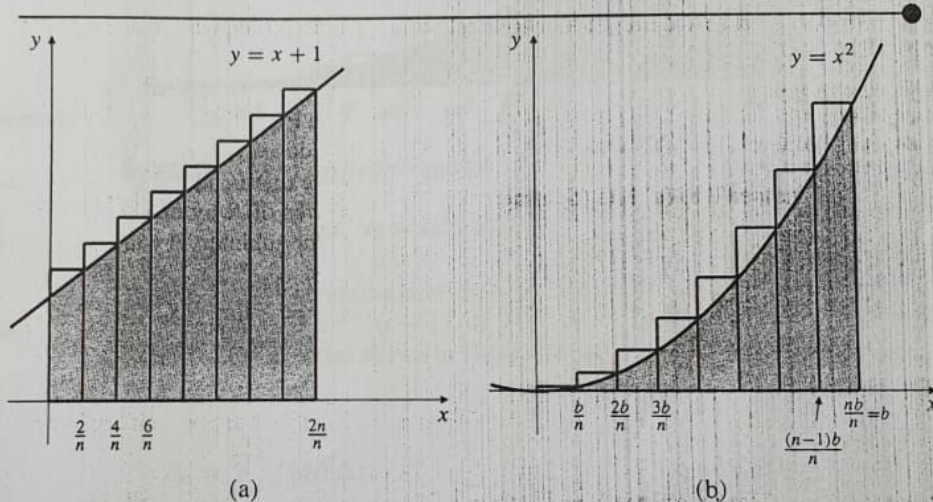


Figure 5.7

- (a) The region of Example 1
(b) The region of Example 2

Solve →

EXAMPLE 2

Find the area of the region bounded by the parabola $y = x^2$ and the straight lines $y = 0$, $x = 0$, and $x = b$, where $b > 0$.

Solution The area A of the region is the limit of the sum S_n of areas of the rectangles shown in Figure 5.7(b). Again we have used equal subintervals, each of length b/n . The height of the i th rectangle is $(ib/n)^2$. Thus,

$$S_n = \sum_{i=1}^n \left(\frac{ib}{n} \right)^2 \frac{b}{n} = \frac{b^3}{n^3} \sum_{i=1}^n i^2 = \frac{b^3}{n^3} \frac{n(n+1)(2n+1)}{6},$$

by formula (c) of Theorem 1. Hence, the required area is

$$A = \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} b^3 \frac{(n+1)(2n+1)}{6n^2} = \frac{b^3}{3} \text{ square units.}$$

Finding an area under the graph of $y = x^k$ over an interval I becomes more and more difficult as k increases if we continue to try to subdivide I into subintervals of equal length. (See Exercise 14 at the end of this section for the case $k = 3$.) It is, however, possible to find the area for arbitrary k if we subdivide the interval I into subintervals whose lengths increase in geometric progression. Example 3 illustrates this.

EXAMPLE 3

Let $b > a > 0$, and let k be any real number except -1 . Show that the area A of the region bounded by $y = x^k$, $y = 0$, $x = a$, and $x = b$ is

$$A = \frac{b^{k+1} - a^{k+1}}{k+1} \text{ square units.}$$

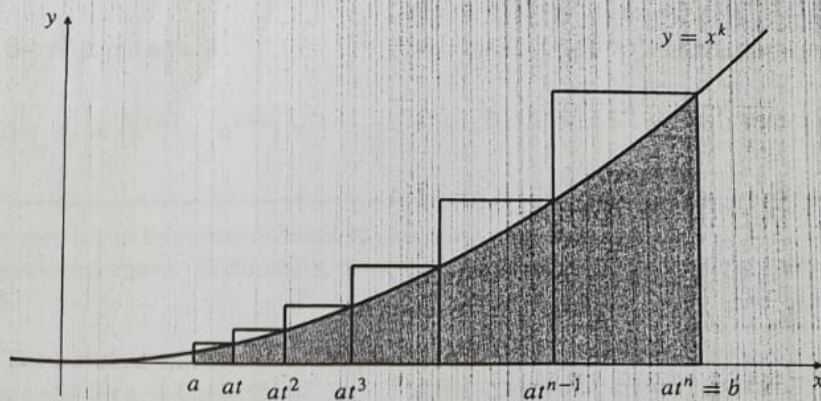


Figure 5.8 For this partition the subinterval lengths increase exponentially

BEWARE! This is a long and rather difficult example. Either skip over it or take your time and check each step carefully.

Solution Let $t = (b/a)^{1/n}$ and let

$$x_0 = a, x_1 = at, x_2 = at^2, x_3 = at^3, \dots, x_n = at^n = b.$$

These points subdivide the interval $[a, b]$ into n subintervals of which the i th, $[x_{i-1}, x_i]$, has length $\Delta x_i = at^{i-1}(t - 1)$. If $f(x) = x^k$, then $f(x_i) = a^k t^{ki}$. The sum of the areas of the rectangles shown in Figure 5.8 is:

$$\begin{aligned} S_n &= \sum_{i=1}^n f(x_i) \Delta x_i \\ &= \sum_{i=1}^n a^k t^{ki} at^{i-1}(t - 1) \\ &= a^{k+1} (t - 1) t^k \sum_{i=1}^n t^{(k+1)(i-1)} \\ &= a^{k+1} (t - 1) t^k \sum_{i=1}^n r^{(i-1)} \quad \text{where } r = t^{k+1} \\ &= a^{k+1} (t - 1) t^k \frac{r^n - 1}{r - 1} \quad (\text{by Theorem 1(d)}) \\ &= a^{k+1} (t - 1) t^k \frac{t^{(k+1)n} - 1}{t^{k+1} - 1}. \end{aligned}$$

Now replace t with its value $(b/a)^{1/n}$ and rearrange factors to obtain

$$\begin{aligned} S_n &= a^{k+1} \left(\left(\frac{b}{a} \right)^{1/n} - 1 \right) \left(\frac{b}{a} \right)^{k/n} \frac{\left(\frac{b}{a} \right)^{k+1} - 1}{\left(\frac{b}{a} \right)^{(k+1)/n} - 1} \\ &= (b^{k+1} - a^{k+1}) c^{k/n} \frac{c^{1/n} - 1}{c^{(k+1)/n} - 1}, \quad \text{where } c = \frac{b}{a}. \end{aligned}$$

Of the three factors in the final line above, the first does not depend on n , and the second, $c^{k/n}$, approaches $c^0 = 1$ as $n \rightarrow \infty$. The third factor is an indeterminate form of type $[0/0]$, which we evaluate using l'Hôpital's Rule. First let $u = 1/n$. Then

$$\begin{aligned}\lim_{n \rightarrow \infty} \frac{c^{1/n} - 1}{c^{(k+1)/n} - 1} &= \lim_{u \rightarrow 0^+} \frac{c^u - 1}{c^{(k+1)u} - 1} \quad \left[\frac{0}{0} \right] \\ &= \lim_{u \rightarrow 0^+} \frac{c^u \ln c}{(k+1)c^{(k+1)u} \ln c} = \frac{1}{k+1}.\end{aligned}$$

Therefore, the required area is

$$A = \lim_{n \rightarrow \infty} S_n = (b^{k+1} - a^{k+1}) \times 1 \times \frac{1}{k+1} = \frac{b^{k+1} - a^{k+1}}{k+1} \text{ square units.}$$

As you can see, it can be rather difficult to calculate areas bounded by curves by the methods developed above. Fortunately, there is an easier way, as we will discover in Section 5.5.

Remark For technical reasons it was necessary to assume $a > 0$ in Example 3. The result is also valid for $a = 0$ provided $k > -1$. In this case we have $\lim_{a \rightarrow 0^+} a^{k+1} = 0$, so the area under $y = x^k$, above $y = 0$, between $x = 0$ and $x = b > 0$, is $A = b^{k+1}/(k+1)$ square units. For $k = 2$ this agrees with the result of Example 2.

EXAMPLE 4 Identify the limit $L = \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{n-i}{n^2}$ as an area, and evaluate it.

Solution We can rewrite the i th term of the sum so that it depends on i/n :

$$L = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left(1 - \frac{i}{n}\right) \frac{1}{n}.$$

The terms now appear to be the areas of rectangles of base $1/n$ and heights $1 - x_i$, ($1 \leq i \leq n$), where

$$x_1 = \frac{1}{n}, \quad x_2 = \frac{2}{n}, \quad x_3 = \frac{3}{n}, \quad \dots, \quad x_n = \frac{n}{n}.$$

Thus, the limit L is the area under the curve $y = 1 - x$ from $x = 0$ to $x = 1$. (See Figure 5.9.) This region is a triangle having area $1/2$ square unit, so $L = 1/2$.

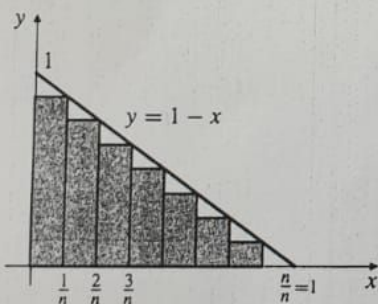


Figure 5.9 Recognizing a sum of areas

EXERCISES 5.2

Use the techniques of Examples 1 and 2 (with subintervals of equal length) to find the areas of the regions specified in Exercises 1–13.

- Below $y = 3x$, above $y = 0$, from $x = 0$ to $x = 1$.
- Below $y = 2x + 1$, above $y = 0$, from $x = 0$ to $x = 3$.
- Below $y = 2x - 1$, above $y = 0$, from $x = 1$ to $x = 3$.
- Below $y = 3x + 4$, above $y = 0$, from $x = -1$ to $x = 2$.
- Below $y = x^2$, above $y = 0$, from $x = 1$ to $x = 3$.
- Below $y = x^2 + 1$, above $y = 0$, from $x = 0$ to $x = a > 0$.
- Below $y = x^2 + 2x + 3$, above $y = 0$, from $x = -1$ to $x = 2$.

- Above $y = x^2 - 1$, below $y = 0$.
- Above $y = 1 - x$, below $y = 0$, from $x = 2$ to $x = 4$.
- Above $y = x^2 - 2x$, below $y = 0$.
- Below $y = 4x - x^2 + 1$, above $y = 1$.

- Below $y = e^x$, above $y = 0$, from $x = 0$ to $x = b > 0$.
- Below $y = 2^x$, above $y = 0$, from $x = -1$ to $x = 1$.

- Use the formula $\sum_{i=1}^n i^3 = n^2(n+1)^2/4$; from Exercises 39–41 of Section 5.1, to find the area of the region lying under $y = x^3$, above the x -axis, and between the vertical lines at $x = 0$ and $x = b > 0$.

- Use the subdivision of $[a, b]$ given in Example 3 to find the

area under $y = 1/x$, above $y = 0$ from $x = a > 0$ to $x = b > a$. Why should your answer not be surprising?

In Exercises 16–19, interpret the given sum S_n as a sum of areas of rectangles approximating the area of a certain region in the plane and hence evaluate $\lim_{n \rightarrow \infty} S_n$.

$$16. S_n = \sum_{i=1}^n \frac{2}{n} \left(1 - \frac{i}{n} \right) \quad 17. S_n = \sum_{i=1}^n \frac{2}{n} \left(1 - \frac{2i}{n} \right)$$

$$18. S_n = \sum_{i=1}^n \frac{2n + 3i}{n^2} \quad 19. S_n = \sum_{j=1}^n \frac{1}{n} \sqrt{1 - (j/n)^2}$$

5.3 The Definite Integral

In this section we generalize and make more precise the procedure used for finding areas developed in Section 5.2, and we use it to define the *definite integral* of a function f on an interval I . Let us assume, for the time being, that $f(x)$ is defined and continuous on the closed, finite interval $[a, b]$. We no longer assume that the values of f are nonnegative.

Partitions and Riemann Sums

Let P be a finite set of points arranged in order between a and b on the real line, say

$$P = \{x_0, x_1, x_2, x_3, \dots, x_{n-1}, x_n\},$$

where $a = x_0 < x_1 < x_2 < x_3 < \dots < x_{n-1} < x_n = b$. Such a set P is called a **partition** of $[a, b]$; it divides $[a, b]$ into n subintervals of which the i th is $[x_{i-1}, x_i]$. We call these the subintervals of the partition P . The number n depends on the particular partition, so we write $n = n(P)$. The length of the i th subinterval of P is

$$\Delta x_i = x_i - x_{i-1}, \quad (\text{for } 1 \leq i \leq n)$$

and we call the greatest of these numbers Δx_i the **norm** of the partition P and denote it $\|P\|$:

$$\|P\| = \max_{1 \leq i \leq n} \Delta x_i.$$

Since f is continuous on each subinterval $[x_{i-1}, x_i]$ of P , it takes on maximum and minimum values at points of that interval (by Theorem 8 of Section 1.4). Thus there are numbers l_i and u_i in $[x_{i-1}, x_i]$ such that

$$f(l_i) \leq f(x) \leq f(u_i) \quad \text{whenever } x_{i-1} \leq x \leq x_i.$$

If $f(x) \geq 0$ on $[a, b]$, then $f(l_i) \Delta x_i$ and $f(u_i) \Delta x_i$ represent the areas of rectangles having the interval $[x_{i-1}, x_i]$ on the x -axis as base, and having tops passing through the lowest and highest points, respectively, on the graph of f on that interval. (See Figure 5.10.) If A_i is that part of the area under $y = f(x)$ and above the x -axis that lies in the vertical strip between $x = x_{i-1}$ and $x = x_i$, then

$$f(l_i) \Delta x_i \leq A_i \leq f(u_i) \Delta x_i.$$

If f can have negative values, then one or both of $f(l_i) \Delta x_i$ and $f(u_i) \Delta x_i$ can be negative and will then represent the negative of the area of a rectangle lying below the x -axis. In any event, we always have $f(l_i) \Delta x_i \leq f(u_i) \Delta x_i$.

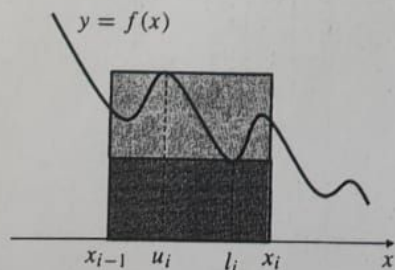


Figure 5.10

DEFINITION

2

Upper and lower Riemann sums

The lower (Riemann) sum, $L(f, P)$, and the upper (Riemann) sum, $U(f, P)$, for the function f and the partition P are defined by:

$$\begin{aligned} L(f, P) &= f(l_1) \Delta x_1 + f(l_2) \Delta x_2 + \cdots + f(l_n) \Delta x_n \\ &= \sum_{i=1}^n f(l_i) \Delta x_i, \\ U(f, P) &= f(u_1) \Delta x_1 + f(u_2) \Delta x_2 + \cdots + f(u_n) \Delta x_n \\ &= \sum_{i=1}^n f(u_i) \Delta x_i. \end{aligned}$$

Figure 5.11 illustrates these Riemann sums as sums of *signed* areas of rectangles; any such areas that lie below the x -axis are counted as negative.

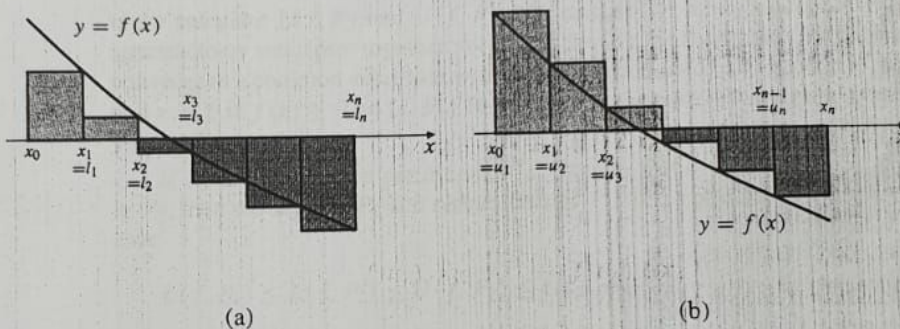


Figure 5.11 (a) A lower Riemann sum and (b) an upper Riemann sum for a decreasing function f . The areas of rectangles shaded in colour are counted as positive; those shaded in grey are counted as negative

EXAMPLE 1

Calculate lower and upper Riemann sums for the function $f(x) = 1/x$ on the interval $[1, 2]$, corresponding to the partition P of $[1, 2]$ into four subintervals of equal length.

Solution The partition P consists of the points $x_0 = 1$, $x_1 = 5/4$, $x_2 = 3/2$, $x_3 = 7/4$, and $x_4 = 2$. Since $1/x$ is decreasing on $[1, 2]$, its minimum and maximum values on the i th subinterval $[x_{i-1}, x_i]$ are $1/x_i$ and $1/x_{i-1}$, respectively. Thus, the lower and upper Riemann sums are

$$L(f, P) = \frac{1}{4} \left(\frac{4}{5} + \frac{2}{3} + \frac{4}{7} + \frac{1}{2} \right) = \frac{533}{840} \approx 0.6345,$$

$$U(f, P) = \frac{1}{4} \left(1 + \frac{4}{5} + \frac{2}{3} + \frac{4}{7} \right) = \frac{319}{420} \approx 0.7595.$$

EXAMPLE 2

Calculate the lower and upper Riemann sums for the function $f(x) = x^2$ on the interval $[0, a]$ (where $a > 0$), corresponding to the partition P_n of $[0, a]$ into n subintervals of equal length.

Solution Each subinterval of P_n has length $\Delta x = a/n$, and the division points are given by $x_i = ia/n$ for $i = 0, 1, 2, \dots, n$. Since x^2 is increasing on $[0, a]$, its minimum and maximum values over the i th subinterval $[x_{i-1}, x_i]$ occur at $l_i = x_{i-1}$ and $u_i = x_i$.

respectively. Thus, the lower Riemann sum of f for P_n is

$$\begin{aligned} L(f, P_n) &= \sum_{i=1}^n (x_{i-1})^2 \Delta x = \frac{a^3}{n^3} \sum_{i=1}^n (i-1)^2 \\ &= \frac{a^3}{n^3} \sum_{j=0}^{n-1} j^2 = \frac{a^3}{n^3} \frac{(n-1)n(2(n-1)+1)}{6} = \frac{(n-1)(2n-1)a^3}{6n^2}, \end{aligned}$$

where we have used Theorem 1(c) of Section 5.1 to evaluate the sum of squares. Similarly, the upper Riemann sum is

$$\begin{aligned} U(f, P_n) &= \sum_{i=1}^n (x_i)^2 \Delta x \\ &= \frac{a^3}{n^3} \sum_{i=1}^n i^2 = \frac{a^3}{n^3} \frac{n(n+1)(2n+1)}{6} = \frac{(n+1)(2n+1)a^3}{6n^2}. \end{aligned}$$

The Definite Integral

If we calculate $L(f, P)$ and $U(f, P)$ for partitions P having more and more points spaced closer and closer together, we expect that, in the limit, these Riemann sums will converge to a common value that will be the area bounded by $y = f(x)$, $y = 0$, $x = a$, and $x = b$ if $f(x) \geq 0$ on $[a, b]$. This is indeed the case, but we cannot fully prove it yet.

If P_1 and P_2 are two partitions of $[a, b]$ such that every point of P_1 also belongs to P_2 , then we say that P_2 is a **refinement** of P_1 . It is not difficult to show that in this case

$$L(f, P_1) \leq L(f, P_2) \leq U(f, P_2) \leq U(f, P_1);$$

adding more points to a partition increases the lower sum and decreases the upper sum. (See Exercise 18 at the end of this section.) Given any two partitions, P_1 and P_2 , we can form their **common refinement** P , which consists of all of the points of P_1 and P_2 . Thus,

$$L(f, P_1) \leq L(f, P) \leq U(f, P) \leq U(f, P_2).$$

Hence, every lower sum is less than or equal to every upper sum. Since the real numbers are complete, there must exist *at least one* real number I such that

$$L(f, P) \leq I \leq U(f, P), \quad \text{for every partition } P.$$

If there is *only one* such number, we will call it the definite integral of f on $[a, b]$.

DEFINITION

3

The definite integral

Suppose there is exactly one number I such that for every partition P of $[a, b]$ we have

$$L(f, P) \leq I \leq U(f, P).$$

Then we say that the function f is **integrable** on $[a, b]$, and we call I the **definite integral** of f on $[a, b]$. The definite integral is denoted by the symbol

$$I = \int_a^b f(x) dx.$$

The definite integral of $f(x)$ over $[a, b]$ is a **number**; it is not a function of x . It depends on the numbers a and b and on the particular function f , but not on the variable x (which is a **dummy variable** like the variable i in the sum $\sum_{i=1}^n f(i)$). Replacing x with another variable does not change the value of the integral:

$$\int_a^b f(x) dx = \int_a^b f(t) dt.$$

The various parts of the symbol $\int_a^b f(x) dx$ have their own names:

- (i) $\int$ is called the **integral sign**; it resembles the letter S since it represents the limit of a sum.
- (ii) a and b are called the **limits of integration**; a is the **lower limit**, b is the **upper limit**.
- (iii) The function f is the **integrand**; x is the **variable of integration**.
- (iv) dx is the **differential** of x . It replaces Δx in the Riemann sums. If an integrand depends on more than one variable, the differential tells you which one is the variable of integration.

EXAMPLE 3

Show that $f(x) = x^2$ is integrable over the interval $[0, a]$, where $a > 0$, and evaluate $\int_0^a x^2 dx$.

Solution We evaluate the limits as $n \rightarrow \infty$ of the lower and upper sums of f over $[0, a]$ obtained in Example 2 above.

$$\begin{aligned}\lim_{n \rightarrow \infty} L(f, P_n) &= \lim_{n \rightarrow \infty} \frac{(n-1)(2n-1)a^3}{6n^2} = \frac{a^3}{3}, \\ \lim_{n \rightarrow \infty} U(f, P_n) &= \lim_{n \rightarrow \infty} \frac{(n+1)(2n+1)a^3}{6n^2} = \frac{a^3}{3}.\end{aligned}$$

If $L(f, P_n) \leq I \leq U(f, P_n)$, we must have $I = a^3/3$. Thus, $f(x) = x^2$ is integrable over $[0, a]$, and

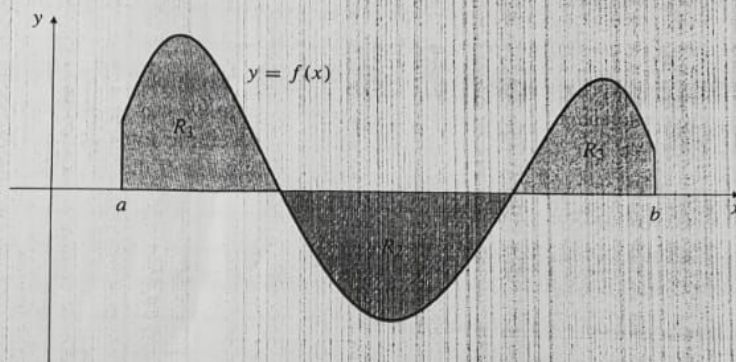
$$\int_0^a f(x) dx = \int_0^a x^2 dx = \frac{a^3}{3}.$$

For all partitions P of $[a, b]$, we have

$$L(f, P) \leq \int_a^b f(x) dx \leq U(f, P).$$

If $f(x) \geq 0$ on $[a, b]$, then the area of the region R bounded by the graph of $y = f(x)$, the x -axis, and the lines $x = a$ and $x = b$ is A square units, where $A = \int_a^b f(x) dx$. If $f(x) \leq 0$ on $[a, b]$, the area of R is $-\int_a^b f(x) dx$ square units. For general f , $\int_a^b f(x) dx$ is the area of that part of R lying above the x -axis minus the area of that part lying below the x -axis. (See Figure 5.12.) You can think of $\int_a^b f(x) dx$ as a “sum” of “areas” of infinitely many rectangles with heights $f(x)$ and “infinitesimally small widths” dx ; it is a limit of the upper and lower Riemann sums.

Figure 5.12 $\int_a^b f(x) dx$ equals area R_1 - area R_2 + area R_3



General Riemann Sums

Let $P = \{x_0, x_1, x_2, \dots, x_n\}$, where $a = x_0 < x_1 < x_2 < \dots < x_n = b$, be a partition of $[a, b]$ having norm $\|P\| = \max_{1 \leq i \leq n} \Delta x_i$. In each subinterval $[x_{i-1}, x_i]$ of P pick a point c_i (called a *tag*). Let $c = \{c_1, c_2, \dots, c_n\}$ denote the set of these tags. The sum

$$\begin{aligned} R(f, P, c) &= \sum_{i=1}^n f(c_i) \Delta x_i \\ &= f(c_1) \Delta x_1 + f(c_2) \Delta x_2 + f(c_3) \Delta x_3 + \dots + f(c_n) \Delta x_n \end{aligned}$$

is called the **Riemann sum** of f on $[a, b]$ corresponding to partition P and tags c .

Note in Figure 5.13 that $R(f, P, c)$ is a sum of *signed* areas of rectangles between the x -axis and the curve $y = f(x)$. For any choice of the tags c , the Riemann sum $R(f, P, c)$ satisfies

$$L(f, P) \leq R(f, P, c) \leq U(f, P).$$

Therefore, if f is integrable on $[a, b]$, then its integral is the limit of such Riemann sums, where the limit is taken as the number $n(P)$ of subintervals of P increases to infinity in such a way that the lengths of all the subintervals approach zero. That is,

$$\lim_{\substack{n(P) \rightarrow \infty \\ \|P\| \rightarrow 0}} R(f, P, c) = \int_a^b f(x) dx.$$

As we will see in Chapter 7, many applications of integration depend on recognizing that a limit of Riemann sums is a definite integral.

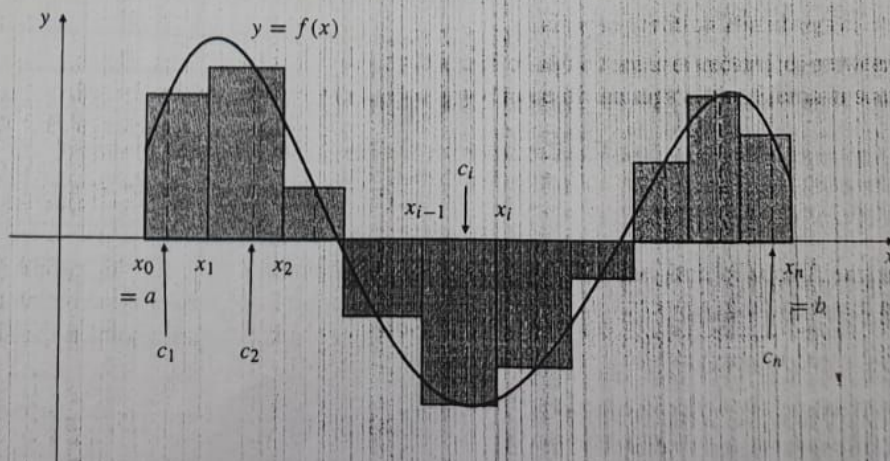


Figure 5.13 The Riemann sum $R(f, P, c)$ is the sum of areas of the rectangles shaded in colour minus the sum of the areas of the rectangles shaded in grey